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Medicine

Objectives

1. Basic concepts of Rheumatology
2. Common Pediatric Pathologies
 1. Juvenile Idiopathic Arthritis (JIA)
 2. Spondyloarthritides (Juvenile Ankylosing Spondyloarthritis, Psoriatic Arthritis, Reactive Arthritis)
 3. Pediatric SLE (pSLE)
 4. Neonatal Lupus
 5. Juvenile Dermatomyositis (JDM)
 6. Bechet disease (BD)
 7. Juvenile Sjogren's Disease (JSD)
 8. Pediatric Scleroderma
 9. FMF
3. Resources
4. Summarization

Basic concepts

Rheum disease = History + Physical exam findings + autoimmune markers + serologic tests + pathology + imaging.

What clues you in that your patient may have a Rheumatologic condition?

Fevers

Arthralgias (joint pain) WITH arthritis (inflammation of the joint)

Timing (Morning stiffness – gets better as you use it more)

Rash

Weakness/Fatigue

Duration of symptoms

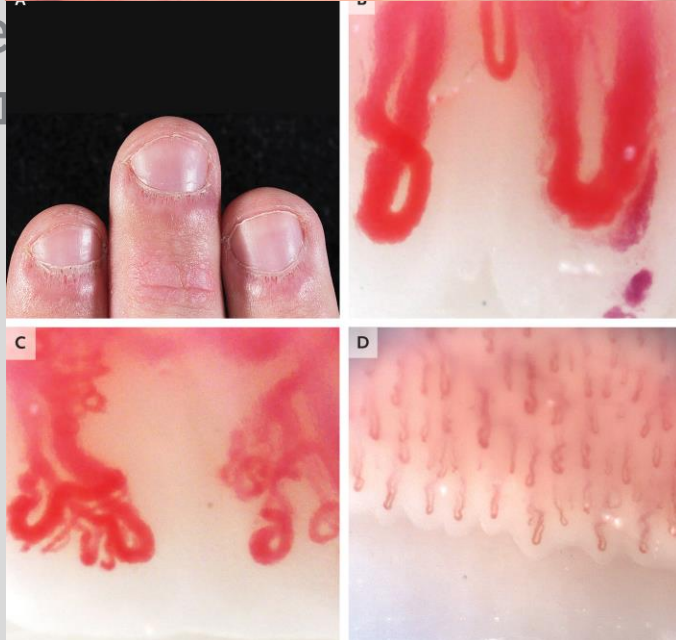
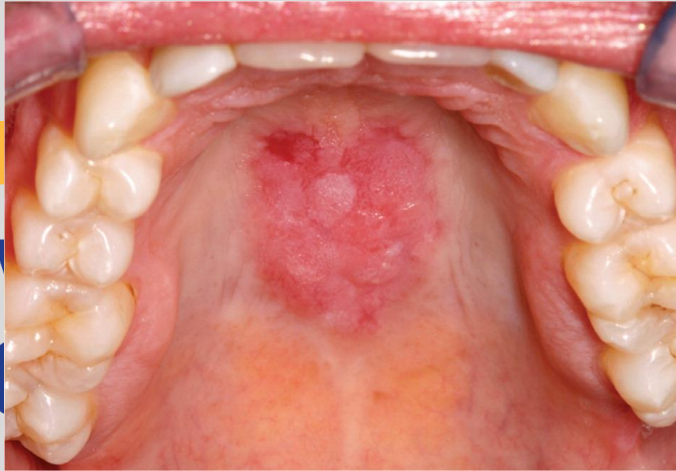
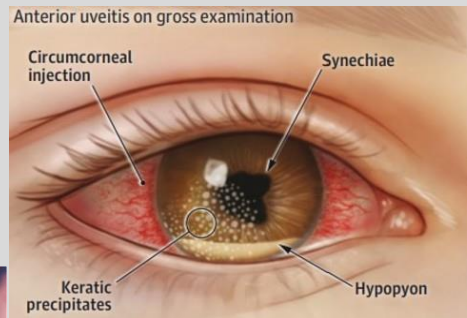
What you don't want to miss

Cancer (ALL/neuroblastoma)

Infection (viral illness, septic arthritis, lyme, osteomyelitis)

Mimics: metabolic (thyroid), orthopedic (patellofemoral syndrome, growing pains, chronic pain conditions (fibromyalgia, depression), other autoimmune disorders (IBD), vasculitidis

Basic concepts



For diagnosis: complete Physical Exam (as disease is multisystem)

Thorough Skin exam + scalp (generalized skin rash, sparing nasal labial folds, involving ears/around eyes, lower extremities, hair patterns)

Nail Beds (dilated capillary loops) - Nailfold capillaroscopy

Oropharynx: ulcers (hard palate involved in SLE), mucositis, TMJ joint instability/inflammation

Lymph nodes: lymphadenopathy

Eyes: uveitis, conjunctivitis

Heart: pericardial rubs

Lungs: pleural effusions, ILD

All Joints: Arthritis defined as intra-articular swelling of 2 or more of the following findings (pain on motion, range of motion limitation and warmth), limb length discrepancy



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Imaging: Plain xrays

MRI more sensitive for early
erosive arthritis

Echo

PFTs

CT chest

Basic labs:

Lymphopenia more specific to SLE than Leukopenia, in sJIA WBC Elevated
PLTs elevated – due to inflammation

Hbg low – anemia of chronic disease – nonspecific

ESR, CRP elevation– non specific

Rheumatoid Factor + is present in <10% of children with JIA – poor
sensitivity. Co-relate with symptoms = strong prognostic indicator

Anti- cyclic citrullinated protein (CCP) = RF pos JIA pretty specific

Anti- nuclear antibody – low specificity – not recommended for screening
but correlate with symptoms

DsDNA = SLE

Anti-Smith = SLE

SSA (Ro) positive = SLE, sjogren syndrome

SSA (La) positive = SLE, sjogren syndrome

Anti-histone = Drug induced lupus

Anti-neutrophil cytoplasmic antibodies (ANCA) = vasculitis

Anti- topoisomerase (Scl-70) = systemic sclerosis – rare in childhood

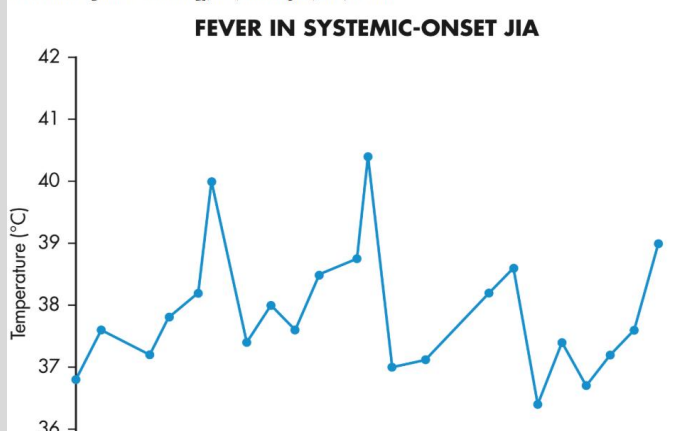
Muscle Enzymes – ALT, AST, CK, Aldolase, LDH

Urine – protein, can do protein/creatinine ratio

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From Hochberg MC: *Rheumatology*, ed 7, Philadelphia, 2018, Elsevier.



Common Pathologies

JIA (Juvenile Idiopathic Arthritis)



Most common rheumatic disease in children, clinical diagnosis

Age at onset: < 16 years old

Arthritis: > or = 1 joint

Duration: > or = 6 weeks

Type of articular involvement - many subtypes :

Poly: > or = 5 joints

RF neg or RF pos

Oligo: < or = 4 joints

Systemic-onset disease: arthritis with rash and quotidian fever

Psoriatic arthritis

Enthesitis-related arthritis



UTE



FIG. 196.9
CT scan of the temporomandibular joint of a child with juvenile idiopathic arthritis exhibiting destruction on the right.

Joints affected:

Oligo: larger joints

Polyarthritis: both upper, lower joints. Can also affect TMJs, cervical spine involvement and hip disease

Enthesitis – lower limb joints

Treatment:

NSAIDs (Naproxen, Ibuprofen, Meloxicam)

DMARDs (Methotrexate PO or SC weekly)

Biologics (Anti-TNF-alpha, anti-IL1 receptor)

Intra-articular corticosteroids

Systemic steroids – generally avoided unless bridging or severe acute disease



FIG. 196.11
Severe hip disease in 13-yr-old male with active systemic juvenile idiopathic arthritis. Radiograph shows destruction of the femoral head and acetabula, joint space narrowing, and subluxation of left hip. The child had received corticosteroids systemically for 9 years.



FIG. 7.7

Systemic juvenile idiopathic arthritis. (A and B) The rash is erythematous, macular, and often evanescent. It can be more prominent during periods of fever. Featured on trunk and extremities.

Prognosis:

Goal is disease remission

40-60% of patients get better with 6 weeks of NSAIDs

Oligo fares better than Poly

Depends on disease severity, earlier age at presentation, RF seropositive, disease involving hip, hand, wrist generally have poorer prognosis

Complications include leg discrepancy, micrognathia, contractures, persistent uncontrolled uveitis can cause chronic scarring and blindness

Anatomic region	Enthesitis exam
Foot and ankle	Achilles tendon insertion to calcaneus Plantar fascia insertion to calcaneus Plantar fascia insertion to metatarsal heads Plantar fascia insertion to base of 1st metatarsal
Knee	Quadriceps tendon insertion to patella (2 and 10 o'clock) Infrapatellar ligament insertion to patella (6 o'clock) and tibial tuberosity
Pelvis	Hip extensor insertion at greater trochanter of femur Sartorius insertion at anterior superior iliac spine Posterior superior iliac spine Abdominal muscle insertions to iliac crest Gracilis and adduction insertion to pubis symphysis Hamstring insertion to ischial tuberosity
Spine	5th lumbar spinous process
Upper extremity	Common flexor insertion at medial epicondyle of humerus Common extensor insertion at lateral epicondyle of humerus Supraspinatus insertion into greater tuberosity of humerus
Chest	Costovertebral junctions (1st and 7th)



FIG. 197.2

Anatomic sites for assessment of enthesitis in ERA and JAS.

Enthesitis related arthritis (ERA)

Classification Criteria

1. Arthritis PLUS Enthesitis
2. Arthritis OR Enthesitis with 2 of more of the following
 - Sacroiliac joint tenderness or inflammatory lumbosacral pain
 - Presence of HLA-B27
 - Onset of arthritis in a male older than 6 years
 - Acute anterior Uveitis
 - Family history of HLA-B27 associated disease in 1st degree relative
3. Typically first 6 months of disease are asymmetric and have fewer than 4 joints affected (inflammation of small joints in foot – highly suggestive of ERA)



FIG. 197.3
Loss of lumbodorsal spine mobility in a boy with ankylosing spondylitis. The lower spine remains straight when the patient bends forward.

Juvenile Ankylosing Spondylitis (JAS)

Frequently begins with oligoarthritis and enthesitis

Classification Criteria (modified New York criteria)

1. Arthritis PLUS Enthesitis
2. Arthritis OR Enthesitis with 2 of more of the following
 - Sacroiliac joint tenderness or inflammatory lumbosacral pain
 - Presence of HLA-B27 (90% of children with JAS)
 - Onset of arthritis in a male older than 6 years
 - Acute anterior Uveitis
 - Family history of HLA-B27 associated disease in 1st degree relative
3. Typically first 6 months of disease are asymmetric and have fewer than 4 joints affected
4. Treatment: goals of therapy to control inflammation, prevent ankylosis (fusion of bones) with DMARDS, NSAIDS, Physical Therapy
5. Prognosis: if ongoing disease activity for >5 yrs = disability very likely , remission in < 20% of those patients

Symptoms Characteristic of Inflammatory Back Pain

Pain at night with morning stiffness (and improvement on arising)

No improvement with rest

Improvement with exercise

Insidious onset

Good response to nonsteroidal antiinflammatory drugs

Reactive Arthritis

Quick review

Defined as joint inflammation caused by a sterile inflammatory reaction after recent infection 3d to 6wks

Can't see, can't pee, can't climb a tree – **not** common in kids. Usually more systemic symptoms (fatigue, weakness), cutaneous manifestations (erythematous oral macules, plaques, erosions, erythema nodosum, fingertip erosions and more enthesitis and dactylitis)

Arthritis is asymmetric, oligoarticular of lower extremities.

Bacteria: Salmonella, Shigella, Yersinia, Campylobacter, GU tract infections with Chlamydia, Ecoli and C diff less common

Virus: Adeno, CMV, EBV, HSV, Parvovirus, Rubella

75% of pts with reactive arthritis are HLA-B27 positive. Tx NSAIDS

If Chlamydia ongoing = TREAT the bug



FIG. 198.1

Enthesitis—swelling of the posterior aspect of the left heel and lateral aspect of the ankle.



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Mimics:

Acute Rheumatic Fever: caused by untreated GAS infection, 2-4 wks after illness. JONES criteria, will have migratory polyarthritis

Neisseria Gonococcal arthritis: rash, fever, polyarticular migratory, sexually active. Tx: CTX

Lyme Arthritis: can mimic oligoarticular JIA, arthritis can be a late manifestation months or years after lyme exposure, usually knee, can get titers to help with diagnosis

Pyogenic/Septic Arthritis: acute symptoms of inflammation of one joint, aspiration of joint fluid showed increased WBC with neutrophil predominance



Pediatric Onset SLE (pSLE)

Compared with adults, kids have more severe disease and widespread organ involvement

In adolescents: renal failure or nephrotic syndrome – 1st presenting sign

Prevalence of SLE in children 1-6/100,000 highest among African, Asian, Hispanic, Native American descent. Childhood SLE is rare before 5yr, median age at diagnosis 11-12 years. 20% of SLE pts are diagnosed before age 16yr.

Most common presenting sign in pSLE fever, fatigue, hematologic abnormalities, arthralgias and arthritis (symmetric polyarthritis large and small joints). Neuropsych complications hard on peds patients

Antiphospholipid antibodies found in 66% of pSLE

Tx: hydroxychloroquine, avoid sun exposure, steroids, MTX, biologics

Prognosis: 5yr and 10 yr survival with \$\$ countries 99, 97%, \$ countries 85, 79%



FIG. 7.13



FIG. 199.1

Mucocutaneous manifestations of SLE. A, Malar rash. B, Vasculitic rash on toes. C, Oral mucosal ulcers. D, Discoid rash in malar distribution.

Consensus Classification of Pediatric Behçet Disease

From Koné-Paut I, Shahram F, Darce-Bello M, et al, for PEDBD group. Consensus classification criteria for paediatric Behçet's disease from a prospective observational cohort: PEDBD. *Ann Rheum Dis* . 2016;75:958–964.

ITEM	DESCRIPTION
Recurrent oral aphthosis	At least three attacks/year
Genital ulceration or aphthosis	Typically with scar
Skin involvement	Necrotic folliculitis, acneiform lesions, erythema nodosum
Ocular involvement	Anterior uveitis, posterior uveitis, retinal vasculitis
Neurologic signs	With the exception of isolated headaches
Vascular signs	Venous thrombosis, arterial thrombosis, arterial aneurysm

Behcet disease (BD)

Spectrum of symptoms is broad - vasculitis of arteries + veins

Autoinflammatory nature: polygenic disorder, innate immunity activation

Prevalence in children is 10% of adult prevalence +FH in 20% of cases.

Mean age at first symptom 8-10 years of age.

Starts with painful oral ulcer does not scar, genital ulcers heal with scars.
Anterior Uveitis. Skin lesions like erythema nodosum

Meet **3 of the criteria** in chart

Tx: Azathioprine for eye disease. Topical tx for ulcers, colchicine, sucralfate, steroids. Mortality in children with BD low except if they get pulm aneurysm



FIG. 203.1
T2-weighted MRI of a child with Sjögren syndrome showing parotitis (arrows).

Juvenile SS	
CLINICAL SYMPTOMS	
1.	Recurrent parotitis or parotid enlargement
2.	Recurrent conjunctivitis (nonallergic and noninfectious)
3.	Recurrent vaginitis
4.	Systemic: fever of unknown origin, arthralgias, hypokalemic paralysis, or abdominal pain
OBJECTIVE	
5.	Ocular dryness (ocular staining or a Schirmer test)
6.	Abnormal sialography
7.	Elevated serum amylase
8.	Leukopenia or elevated ESR
9.	Hyperimmunoglobulinemia (polyclonal)
10.	Renal tubular acidosis
SEROLOGY	
11.	At least one of anti-SSA, anti-SSB, high titer ANA (speckled pattern), RF
HISTOPATHOLOGY	
12.	Lymphocytic infiltration of salivary glands or other organs
DIAGNOSIS OR CLASSIFICATION REQUIREMENTS	
13.	Diagnosis requires at least 4 of 12 items

Juvenile Sjogren's Disease

Rare in children, classic signs of dry eyes and dry mouth
 Mean age at diagnosis 9-10 yr. 75% are F
 Diagnostic criteria in children proposed but not validated

Recurrent **parotid gland enlargement** and **parotiditis** common in children (>70%). Children higher rate of systemic sx, fever, adenopathy compared to adults. In kids also dental caries as a result of dry mouth.

Diagnosis based on clinical features, supported by biopsy of salivary or parotid gland -> lymphocytic infiltration (children more likely to have nml salivary and abnormal parotid biopsies) need **4 of 12 items**

Elevated ESR, positive RF, hypergammaglobulinemia, presence of SSA and SSB antibodies

Treatment: Artificial tears, massage parotids, oral lozenges, fluids.
 Steroids, NSAIDS, hydroxychloroquine



FIG. 204.2
Characteristic erysipeloid erythema associated with familial Mediterranean fever. This rash appears during a flare and overlies the ankle or dorsum of the foot.

Diagnostic Criteria for Familial Mediterranean
From Livneh A, Langevitz P, Zemer D, et al. *Crit*
diagnosis of familial Mediterranean fever. *Arth*
1997;40:1879–2045.

MAJOR CRITERIA

1. Typical attacks [†] with peritonitis (generalized)
2. Typical attacks with pleuritis (unilateral) or pericarditis
3. Typical attacks with monoarthritis (hip, knee, ankle)
4. Typical attacks with fever alone
5. Incomplete abdominal attack

MINOR CRITERIA

1. Incomplete attacks [†] involving chest pain
2. Incomplete attacks involving monoarthritis
3. Exertional leg pain
4. Favorable response to colchicine

Familial Mediterranean Fever (FMF)

Autosomal recessive pathogenic variant in MEFV (amino acid protein denoted pyrin (greek for fever), found in granulocytes, monocytes and in fibroblasts. Causing anto-inflammatory disease.

Occurs primarily among Mediterranean ancestry (Jewish, Turkish, Armenian, Arab, Italian)

Short Recurrent (1-3d) fever, serositis, mono or pauciarticular arthritis or rash. Most present with sx in childhood, 90% present before 20 years. So fever + abd pain or chest pain, arthritis and rash = sounds familiar.

Distinguishing factors: in between fever episodes, no symptoms and cutaneous finding erysipeloid erythematous rash on ankle.

Episodes triggered by stress, infections or menses. Inflammatory markers will be elevated.

Frequency of attacks vary from weekly to one to two flares per year

Tx: prophylactic daily oral colchicine decreases frequency, severity intensity of flares



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Summary!

I know it's confusing

Distinguishing factors:

***JIA – younger child, so many types, remember systemic type

Ankylosing Spondylitis – M, SI joint pain, restriction in ROM, HLA-B27

Psoriasis – Nail findings, rash, joint pain

Reactive Arthritis – after infection, joint pain, uveitis, fatigue, skin findings, HLA-B27

pSLE – similar to adult SLE, multiorgan involvement, complement markers decreased

Behcet – genital and oral ulcers, uveitis, vasculitis

JDM – rash + muscle weakness

Sjogren's – parotiditis, eye dryness

Scleroderma – calcinosis + fibrosis – cutaneous or systemic